Patient Graph: Plain-Language Guide

BRCA TCGA PanCancer Atlas 2018 — milestone 1

> **Main idea:** the graph records one patient's clinical context and molecular observations from that patient's tumour sample. It is not yet a complete biological interaction network or a diagnosis model.

1. What is a feature?

A feature is one item of information, such as age, cancer stage, a mutation type, or a copy-number state. Features are usually stored as **attributes** on a node or edge; they are not automatically nodes.

```
Patient: age = 55, sex = Female, subtype = BRCA_LumA
```

We do not create separate nodes for values such as “Age 55” or “Female,” because they describe the patient rather than being independent biological entities.

2. The graph structure

```
Patient --has_sample--> Sample --has_mutation / has_CNA--> Gene
```

Graph item	Meaning	Examples
Patient node	The person enrolled in the study.	`TCGA-A2-A0CO`
Sample node	A tumour specimen; molecular measurements belong here.	`TCGA-A2-A0CO-01`
Gene node	A gene with mutation or CNA evidence in this sample.	`BCL10`, `CLCA1`
Edge	A labelled fact between nodes.	`has_sample`, `has_mutation`, `has_CNA`

3. Features included

Patient clinical context

Dataset field	Plain meaning
`PATIENT_ID`	Study identifier for one person.
`AGE`	Age at diagnosis.
`SEX`	Reported sex.
`CANCER_TYPE_ACRONYM`	`BRCA`, the study's breast invasive carcinoma label.

`SUBTYPE`	Molecular subtype: Luminal A, Luminal B, HER2-enriched, or basal-like, when available.
`AJCC_PATHOLOGIC_TUMOR_STAGE`	Overall pathological cancer stage, such as Stage IIA or Stage IV.
`PATH_M_STAGE`	Distant-metastasis category: M0 means none identified; M1 means identified; MX means not assessable.

Sample context

Dataset field	Plain meaning
`SAMPLE_ID`	Study identifier for the biological specimen.
`ONCOTREE_CODE`	Short standardized code for tumour histology.
`CANCER_TYPE`, `CANCER_TYPE_DETAILED`, `TUMOR_TYPE`	Broad and detailed cancer/pathology descriptions.
`ANEUPLOIDY_SCORE`	Summary of large chromosome-number abnormalities in the tumour.
`SAMPLE_TYPE`	Specimen category, for example a primary tumour.
`mutation_record_count`	Derived raw MAF-row count; not a clinical tumour-mutation-burden value.

4. Mutations and MAF

A **mutation** is a DNA sequence change. The project reads `data_mutations.txt`, a **MAF** (Mutation Annotation Format) table. Each row is one reported mutation record.

Term/field	Plain meaning
`Hugo_Symbol`	Standard short gene name, such as `TP53`.
`Entrez_Gene_Id`	Numerical gene identifier used to reduce name ambiguity.
`Variant_Classification`	Broad mutation effect: missense, nonsense, frameshift, intronic, etc.
`Variant_Type`	DNA change: SNP (one letter changed), insertion (letters added), or deletion (letters removed).
`HGVSp_Short`	Short protein effect, for example `p.Arg175His`.
Chromosome, position, alleles	Where the change was measured and its before/after DNA letter(s).
`IMPACT`, `FILTER`, `NCALLERS`	Broad predicted consequence, quality status, and support from mutation-calling software.

A raw MAF-row count is not automatically the number of cancer-driving mutations, unique mutated genes, or tumour mutation burden.

5. CNA: copy-number alteration

Humans usually have two copies of most genes. Cancer cells can lose DNA or gain extra DNA copies. A **CNA** is this kind of change.

GISTIC value	Meaning
-2	Homozygous deletion: both copies are estimated lost.
-1	Hemizygous deletion: one copy is estimated lost.
0	Neutral: no discrete gene-level CNA was called.
1	Gain: extra copy/copies are estimated.
2	High-level amplification: strongly increased copy number is estimated.

GISTIC is the dataset's computational method for converting copy-number measurements into these categories. Neutral does **not** mean a gene is unimportant or healthy; it means no CNA was called for that gene/sample.

> **Why neutral CNA calls are excluded:** each sample has about 25,000 genes. If neutral calls became edges, the graph would mostly contain thousands of “no CNA” relationships. We retain -2, -1, 1, and 2 as alteration evidence. Raw zero values remain in the source dataset.

6. Why the example is a star graph

`TCGA-A2-A0CO` has one sample, `TCGA-A2-A0CO-01`. The sample is the source of each molecular fact, so it sits at the centre. This case has zero MAF mutation records and 5,496 retained non-neutral CNA gene calls: 1 Patient + 1 Sample + 5,496 Gene nodes = 5,498 nodes.

```
Patient → Sample → Gene A; Sample → Gene B; Sample → Gene C
```

The PNG intentionally draws only 30 genes. Purple arrows are CNA facts, black is the patient-to-sample link, and no red mutation arrows appear because this sample has no MAF mutation records. It is not yet a gene-interaction network; pathways, gene-gene interactions, diseases, and drugs belong to later milestones.

7. Why no Top-K method was used

We selected *feature types* based on meaning and data quality, not a predictive ranking algorithm. There is no trained diagnosis model, agreed target label, or single valid definition of “most important” yet. A Top-K list could mean frequent genes, largest CNA values, known breast-cancer genes, variance, or model importance.

The 30-gene image cap is only a display limit, not a biological Top-K policy. A future compact graph could explicitly retain high-level CNA events only, known breast-cancer genes, mutation-plus-CNA genes, or a documented Top-K score.

8. Data intentionally not used yet

- Gene expression, methylation, proteomics, phosphoproteomics, and structural variants: separate high-dimensional data types deferred for later.
- Continuous and arm-level CNA: alternate CNA representations that would duplicate evidence.
- Survival fields such as OS, DFS, PFS, and DSS: outcomes excluded to prevent target leakage.
- Race, ethnicity, ancestry, and technical collection metadata: not needed for the initial biological graph without a specific clinical/governance reason.

Generated for the patient-specific graph construction milestone. Source: cBioPortal BRCA TCGA PanCancer Atlas 2018.